Interactive Shooting Game with Score Management System

Group Members:

Syed Huzaifa Farooq (01-135232-097)
Abdullah Javed (01-135232-003)
Saiyab Waheed (01-135232-085)

Department of Computer Science
Bahria University Islamabad, E-8 Campus
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1 Abstract

This project presents an interactive Shooting Game integrated with a Score Management System developed using 8086 Assembly Language in EMU8086. The system allows users to participate in a number guessing based shooting game where targets are generated dynamically. The player attempts to hit the correct target within limited chances, and scores are calculated accordingly. The system also stores and manages scores using arrays and provides features such as score display and sorting. This project demonstrates low-level programming concepts including loops, arrays, stack operations, procedures, macros, and interrupt handling.

2 Introduction

Assembly language plays a vital role in understanding the internal working of computer systems. This project is designed to provide practical exposure to low-level programming using 8086 architecture. The proposed system combines gaming logic with data management, making it more interactive and complex than traditional programs.

3 Problem Statement

Most assembly language programs are limited to basic operations and do not demonstrate the full capability of low-level programming. There is a need to build an interactive system that integrates multiple concepts such as loops, conditions, arrays, and user interaction.

4 Objectives

- To develop an interactive shooting game.
- To implement score management using arrays.
- To apply sorting algorithms.
- To utilize stack operations.
- To demonstrate procedures and macros.

5 Scope of the Project

Includes:

- Menu-driven shooting game
- Score storage and display
- Sorting of scores

Limitations:

- Text-based interface only
- Limited input range

6 Methodology

The project will be developed using EMU8086. A modular approach will be used including game logic, score handling, sorting, and user interaction.

7 System Design

The system consists of:

- Game Module
- Score Module
- Sorting Module
- Control Module

8 Timeline

Task	Duration
Requirement Analysis	1 week
Design	1 week
Development	2 weeks
Testing	1 week

9 Expected Outcomes

A fully functional shooting game demonstrating assembly language concepts with an interactive system.

10 Resources Required

- EMU8086 Emulator
- Personal Computer

11 Conclusion

This project demonstrates how assembly language can be used to build structured and interactive applications while applying core programming concepts.

12 References

- Intel Corporation. *8086 Microprocessor Architecture and Programming Manual.*

- EMU8086 Documentation. *Assembly Language Emulator Guide.*
- Kip R. Irvine, *Assembly Language for x86 Processors*, Pearson Education.
- Class Lecture Notes, Computer Organization and Assembly Language (COAL), Bahria University.